

# Factors Influencing University Students' AI Use and Knowledge Acquisition

Yin, Stella Xin  
Goh, Dion Hoe-Lian

Nanyang Technological University, Singapore | xin013@e.ntu.edu.sg  
Nanyang Technological University, Singapore | ashlgoh@ntu.edu.sg

## ABSTRACT

Despite the growing emphasis on artificial intelligence (AI) education, there is relatively little research on the motivational factors that influence students' intention regarding AI knowledge acquisition and the utilization of AI applications. Understanding these factors helps educators and researchers to develop appropriate interventions to promote AI education. Guided by expectancy-value theory and theory of planned behavior, we examined how university students' beliefs influenced their motivation to learn about and engage with AI technologies. Our findings demonstrated the significant role of expectancy-value beliefs in shaping students' behavioral intention. Additionally, we identified gender differences, which can inform educators in designing gender-specific interventions to enhance female students' motivation in AI learning.

## KEYWORDS

Artificial intelligence, Education, Expectancy-value theory, Theory of planned behavior, Gender differences

## INTRODUCTION

Given the significant impact of AI on education and employment, organizations such as the OECD (2023) and UNESCO (2023) have highlighted the necessity for students to develop AI literacy and competencies in their recent reports. While educational institutions have attempted to support and motivate students to develop AI literacy, there is relatively little research investigates the motivational factors that influence students' intention to acquire AI knowledge and use AI applications (Yin et al., 2023, 2024). Understanding such factors provides valuable insights that educators and researchers can leverage to design effective and engaging AI courses that align with students' needs and expectations.

Expectancy-value theory (EVT) (Eccles, 1983) and theory of planned behavior (TPB) (Ajzen, 2011) are influential theoretical frameworks for understanding factors influencing students' motivation and behaviors. EVT emphasizes how students' learning intention, choices, and effort are influenced by their expectations of success, the perceived value of tasks, and their perceptions of cost. TPB posits that human behavior is determined by behavioral intention, attitudes, and subjective norms. Further, gender is reported to be a significant moderator of behavioral intention and individual beliefs (Venkatesh et al., 2003). These theories have yet to be applied in the context of AI education. Therefore, in this study, we develop a theoretical framework that incorporates these theories to investigate motivational factors that drive university students' AI use and knowledge acquisition. We address two research questions: 1) How do expectancy-value beliefs influence students' behavioral intention to learn about and engage with AI technologies? 2) What are the differences between male and female students in their motivation and behavioral intention?

## METHOD

A total of 141 university students from Singapore (mean age = 23.76, SD = 3.25) participated to this study. There were 66 (46.8%) females and 75 (53.2%) males. In terms of major, 89 (63.1%) students were from STEM subjects such as science and engineering, and 52 (36.9%) students were from non-STEM subjects, such as humanities and social sciences. Participants were required to complete an online survey that measured the following constructs. AI self-efficacy used four items adapted from Chai et al. (2021) ( $\alpha = .88$ ) to evaluate students' beliefs regarding their expectations for success in acquiring AI knowledge and utilizing AI tools. Task value consisted of three components adapted from Nagle (2021): attainment value ( $\alpha = .86$ ), utility value ( $\alpha = .82$ ), and interest value ( $\alpha = .88$ ). Cost was measured by three dimensions adapted from Robinson et al. (2019): effort cost ( $\alpha = .84$ ), emotional cost ( $\alpha = .84$ ), and opportunity cost ( $\alpha = .87$ ). Intention to learn and use AI was adapted from Chai et al. (2020) and Teo (2011) ( $\alpha = .91$ ).

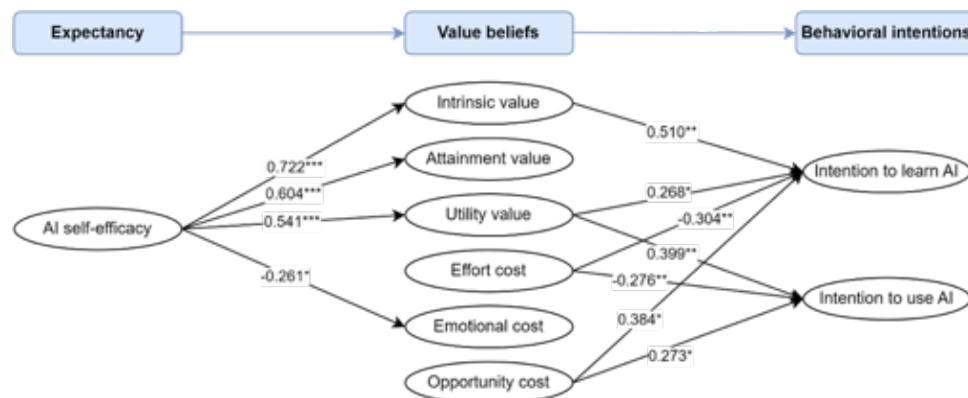
First, confirmatory factor analysis (CFA) was conducted to verify the factor structure of the variables. Second, structural equation modeling (SEM) was constructed to analyze relationships between latent and observed variables. Third, multigroup SEM was used to identify differences between gender groups. All analyses were performed in R software with the R-package lavaan 0.6-17 (Rosseel, 2012).

## FINDINGS

The CFA results showed good fit,  $\chi^2=741.39$ ,  $p < .001$ , CFI = 0.96, TLI = 0.95, RMSEA = 0.05 (95% CI = [0.04, 0.06]), SRMR = 0.05. Next, we constructed a structural model to address the research questions. The model demonstrated a satisfactory fit with CFI = 0.95, TLI = 0.95, RMSEA = 0.06 (95% CI = [0.04, 0.06]), SRMR = 0.07.

Figure 1 depicts the path coefficients of relationships between expectancy-value beliefs and behavioral intention in learning and using AI. AI self-efficacy significantly influenced value beliefs, but it only predicted emotional cost. Students' intrinsic and utility value beliefs positively explained their intention in AI learning and usage. In contrast, the perception of effort cost negatively influenced their intention to learn and use AI. Interestingly, students who perceived higher opportunity cost demonstrated greater behavioral intention to acquire AI knowledge and use AI applications. We also found that intrinsic and utility value played a mediating role between self-efficacy and intention to learn and use AI.

To address RQ2, we examined the differences between gender groups. We observed that the relationship between value beliefs and behavioral intention was more positive among male students than female students. Conversely, the relationship between cost perceptions and behavioral intention was weaker among male students than among females. These findings highlight the need to pay attention to gender-specific interventions to enhance female students' motivation in AI learning and use.



**Figure 1. Relationships among expectancy-value beliefs and behavioral intention.**

## DISCUSSION AND CONCLUSION

Our study supports the significance of value beliefs as key drivers in motivating students' intention to learn and use AI, consistent with previous research (Kelly et al., 2023). The mediation effects of intrinsic value and utility value on self-efficacy and behavioral intention suggest that when individuals perceive themselves as competent to learn or use AI, they are more likely to find that the integration of AI in classrooms is exciting and beneficial. This perception, in turn, enhances their intention to learn and use AI. Surprisingly, the perception of opportunity cost positively predicted intention to learn and use AI. This is probably because individuals' perceptions of opportunity costs are offset by their awareness of the potential benefits they anticipate from engaging with AI learning and technologies. Such awareness might motivate them to invest time and effort into learning and utilizing AI. Finally, gender differences in expectancy-value beliefs and behavior intention have been observed in prior studies in STEM disciplines (Gaspard et al., 2015, 2017). Given that AI is categorized as a STEM subject, it is reasonable that the influence of interest and utility value is lower, while the influence of cost perception is higher among female students.

Our findings advance AI education research and provide a more nuanced understanding of the motivational factors that shape students' behavioral intention in AI learning and use. We extended the EVT and TPB into the AI education context and provide evidence that each dimension of cost perceptions predicted students' behavior intentions individually. Our extended model shows the potential to be applied in other technology-related contexts. Practically, universities could integrate AI-related content into existing curricula and offer more AI-related workshops, to nurture students' intrinsic and utility value beliefs. Furthermore, our study indicates that expectancy-value beliefs have less impact on behavioral intentions among female students compared to their male peers. Therefore, educators should create a more supportive and inclusive learning environment that better accommodates and embraces women students.

Several limitations should be noted. First, self-reported survey data may not fully explain how various factors influence behavioral intention. Second, our participants were recruited from Singapore university, which may limit the generalizability of the findings in other educational contexts. Therefore, future research could employ mixed methods to obtain a fuller understanding of AI perceptions and also include a more diverse demographic.

## GENERATIVE AI USE

We confirm that we did not use generative AI tools/services to author this submission.

## AUTHOR ATTRIBUTION

Yin, Stella Xin: conceptualization, methodology, data curation, formal analysis, writing – original draft; Goh, Dion Hoe-Lian: supervision, writing – review and editing.

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